



# Multi-mode vortex-induced vibration control of long-span bridges by using distributed tuned mass damper inerters (DTMDIs)

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## ABSTRACT

Tuned mass damper inerter (TMDI) has been proved to be a feasible device for vibration control of long-span bridges due to its excellent space performance. However, existing design method for TMDI is conducted mode-by-mode. When applied to multi-mode control of the bridge through multiple TMDIs, it cannot consider the synergistic effect among the TMDIs and thus leading to a suboptimum design result. This paper presents an efficient method for multi-mode vortex-induced vibration (VIV) control of long-span bridges by using spatially distributed TMDIs (DTMDIs). The governing equations of the bridge-DTMDIs system are established in modal coordinate. A global optimization method considering the synergistic effect among the DTMDIs is proposed and compared with conventional mode-by-mode approach through a numerical case study. The robustness of the control schemes against variations in structural and aerodynamic parameters is investigated. The influence of inerter span length on the control efficiency is also discussed. The results show that the proposed global design method can achieve a better control efficiency with even less TMDIs as compared with conventional mode-by-mode approach. The proposed method can also be extended to VIV control of some other line-like structures.

## 1. Introduction

Long-span bridges are sensitive to vortex-induced vibrations (VIVs). There are several bridges that have experienced VIV after erection or during operation, such as the Great Belt East bridge (Larsen et al., 2000), the Xihoumen Bridge (Li et al., 2011), the Trans-Tokyo Bay Bridge (Fujino and Yoshida, 2002), and the very recent Humen Bridge in China (Zheng, 2020). Since VIV threatens the safety of bridge users and fatigue lifetime of the structure, VIV suppression is a major concern for bridge engineers.

Tuned mass damper (TMD) is one of the simplest and most effective devices for vibration control of civil structures. A TMD consists of a mass, a spring, and a dashpot element. The mass element is attached to the main structure through spring and dashpot. With an optimum design of TMD parameters (mass ratio, damping and stiffness), the vibration energy of the main structure is transferred to the TMD and dissipated by the damper (Domaneschi et al., 2015). When applied to bridges, TMD is often installed inside the bridge girder and thus has no influence on the aesthetics of the bridge. It is widely adopted when the aerodynamic control countermeasures are restricted, or as a supplemental plan to

aerodynamic countermeasures to further enhance the control efficiency. Among the prototype bridges that experienced VIV, most of them adopted TMD as the final control scheme, such as the Kessock Bridge (Owen et al., 1996), the Jindo Bridge (Kim et al., 2017), the Rio-Niteroi Bridge (Battista and Pfeil, 2000), the Trans-Tokyo Bay Crossing Bridge (Fujino and Yoshida, 2002), the Volgograd Bridge (Weber and Mašlanka, 2012), and the Chongqi Bridge (Meinhardt et al., 2014).

However, because TMD is installed inside the bridge girder, its feasibility is restricted by the internal space of the girder. The space required by a TMD, as shown in Fig. 1(a), originates from three aspects: (1) the static deformation of TMD springs under gravity, which can be estimated by  $L \approx 0.25/f_n^2$  ( $f_n$  is the natural frequency of the main structure to be controlled) (Meinhardt et al., 2014); (2) the dimension of TMD mass block; (3) the stroke of TMD mass block. The natural frequencies of long-span bridges are quite low. For example, the Xihoumen Bridge experienced 6th, 7th, 9th and 10th vertical mode VIVs, the natural frequencies are 0.18 Hz, 0.23 Hz, 0.27 Hz and 0.32 Hz, respectively (Li et al., 2014). The Great Belt Bridge experienced 3rd, 5th and 6th vertical mode VIVs, the natural frequencies are 0.13 Hz, 0.21 Hz and 0.24 Hz, respectively (Larsen et al., 2000). If the TMD is adopted to

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control these VIV modes, the static deformation of springs could be quite large, thereby failing to be installed inside the bridge girder.

To reduce the space requirement of TMD, researchers proposed to modify the structural form of conventional TMDs. One modification is to use the lever-structure to reduce the static deformation of springs (Li and Zhou, 2004). The other is to use pounding type TMD to reduce the stroke of the mass block (Wang et al., 2018). However, the lever-structure requires large frames for supporting, which significantly increases the total weight and cost of the device. While the pounding type TMD cannot reduce the static deformation of springs, thereby cannot be used to control the low mode vibration of long-span bridges.

Over the past decade or so, a two-terminal device named inerter attracted the attentions of civil engineering community. The fundamental property of this device is to translate the horizontal motion between its two terminals into a high-speed rotation of an object through physical means, such as rack-and-pinion (Smith and Wang, 2004), ball-screw assembly (Papageorgiou and Smith, 2006), and hydraulic mechanism (Wang et al., 2011), etc. Thus, it can generate a resisting force proportional to the relative acceleration between its two terminals. Since the moment of inertia of an object is much larger than its physical mass, the inertial force generated by the inerter can be several orders larger than the one generated by its gravitational mass (Smith, 2002). As a two-terminal inertial element, the inerter can easily be incorporated with conventional passive control systems. According to Ma et al. (2021), the inerter-based passive control systems can be categorized into three groups: (1) inerter-based energy dissipators, such as tuned-viscous-mass-damper (TVMD) (Ikago et al., 2012); (2) inerter-based dynamic vibration absorbers, such as tuned-inerter-damper (TID) (Lazar et al., 2014) and tuned-mass-damper-inerter (TMDI) (Marian and Giaralis, 2014); (3) inerter-based vibration isolators (Cao and Li, 2021; De Domenico and Ricciardi, 2018). Their control efficiencies have been validated through experimental studies (for example, through shaking table test (Pietrosanti et al., 2020)) as well as real applications (for example, NTT East Sendai Aobadori Building (Sugimura et al., 2012)). A comprehensive review on the mechanism and realization of the inerter can be found in Smith (2020). And a state-of-the-art review on the application studies of inerter-based passive control systems can be found in Ma et al. (2021).

The inerter-based vibration isolators are normally placed in between the structure and the ground (or in between upper and lower structures) to shift the fundamental frequency of the structure from the dominate

frequency of external excitations. This device is practically unfeasible to control wind-induced vibration of the bridge, since the wind forces are directly imposed onto the girder. The efficiency of inerter-based energy dissipators for vibration control of long-span bridges is also limited, since it requires relative deformation between its two terminals. However, long-span bridges are horizontal flexible structures. It is hard to place the energy dissipators in between the girder and the ground or in between different positions of the girder to use the relative deformations. Thus, inerter-based dynamic vibration absorbers are the most feasible devices for vibration control of long-span bridges. They can be placed at the maximum response positions of the girder and tuned to the structural frequencies.

Application studies of inerter-based dynamic vibration absorbers (IDVAs) in bridges are quite limited. Jin et al. (2016) firstly investigated the feasibility of inerter-based dynamic vibration absorbers for suppressing beam responses under harmonic excitation. The results showed that inerter-based dynamic vibration absorbers are more efficient than conventional dynamic vibration absorbers, especially for case with a small mass ratio. Xu et al. (2019) adopted tuned mass damper inerter (TMDI) to control the VIV of bridge deck. The tuned mass damper inerter consists of a conventional TMD and an inerter linked in between the TMD mass block and the main deck, as shown in Fig. 1(b). During oscillation, the inertial force of the inerter imposed onto the mass block equals to adding a large virtual mass to the mass block. To be resonant with the main structure, the spring stiffness of the TMDI is tuned by using the sum of the physical mass of the mass block and the virtual mass of the inerter. While, the static stretching of the spring is only generated by the gravity of the mass block. Since the virtual mass of the inerter can be several orders larger than its physical mass, the combination of inerter with conventional TMD can avoid bringing large additional weight to the original structure. The results indicated that the static deformation of TMD springs and the stroke of mass block can be significantly reduced through involving of the inerter. Later, Dai et al. (2019) connected the inerter of TMDI to a different spanwise position of the main deck, and investigated the influence of inerter span length on the control efficiency. Xu et al. (2020) further compared different configurations of inerter-based dynamic vibration absorbers for VIV control of bridges. Dai et al. (2021) proposed to replace the damper element in TMDI by Maxwell element (tandem spring and damper) to suppress VIV of the bridge. All the results indicated that TMDI is a quite feasible device for controlling the low mode vibration of long-span bridges.

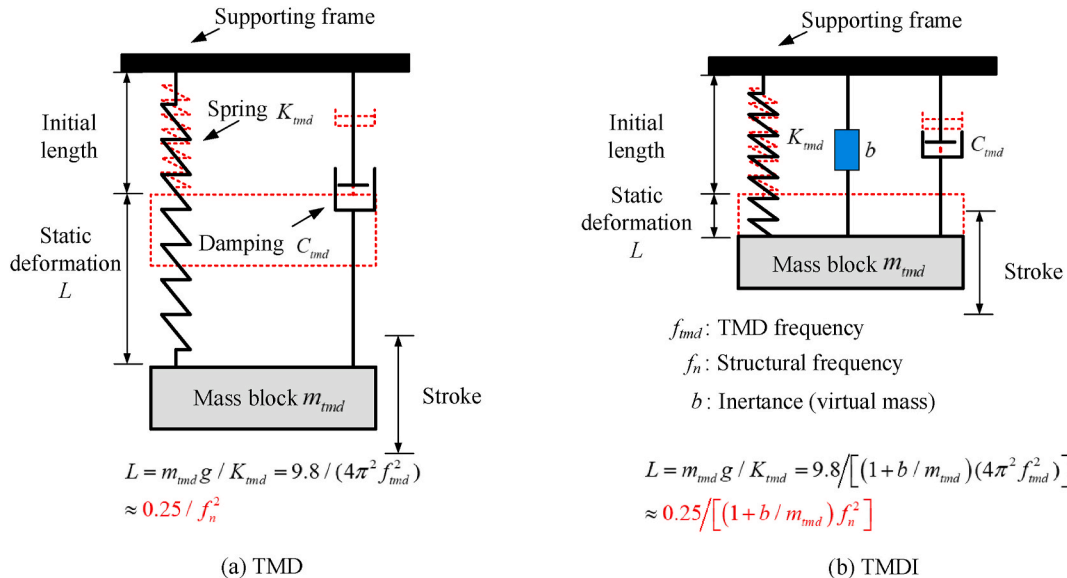


Fig. 1. Diagram of TMD and TMDI.

To get the best control efficiency, one important issue is to optimally design the parameters of the dynamic vibration absorbers. In existing studies, the main structure is generally assumed to be single degree-of-freedom (SDOF), and the excitation is often assumed to be white noise. Based on the fixed-point or norm theory, the design formulas for inerter-based dynamic vibration absorbers can be derived (Hu and Chen, 2015; Marian and Giaralis, 2014). These design formulas neglect the damping of the main structure. Xu et al. (2019) proposed to design the TMDI parameters for bridge VIV control by using numerical optimization algorithm. To consider the nonlinear aeroelastic damping effect of VIV, the response of the girder was iteratively calculated, and the optimum TMDI parameters were found through minimization of the deck's response.

Both the above design procedures assumed the main structure to be SDOF. When applied to multiple degree-of-freedom (MDOF) structures, the inerter-based dynamic vibration absorbers can only be designed mode-by-mode. That is, using one inerter-based dynamic vibration absorber to control one structural mode. And the dynamic property of the main structure adopts the corresponding modal values (Wen et al., 2017). The mode-by-mode approach is easy to implement, but it neglects the synergistic effect among the inerter-based dynamic vibration absorbers. It works well if the modal frequencies of the structure are well separated. But for flexible structures with crowded modal frequencies, the design parameters are suboptimum. To achieve a better control performance, some direct design methods have also been proposed, in which the governing equations of inerter-based dynamic vibration absorbers are incorporated into the structural finite element model (FEM) and numerically optimized by using different objective functions (Radu et al., 2019; Taflanidis et al., 2019).

However, VIV of a structure is dominated by the nonlinear aeroelastic damping effect. Response calculation of the bridge-VIV system also needs iteration. The optimization of multiple inerter-based dynamic vibration absorbers through iterating the finite element model in

physical coordinate can hardly be implemented. In addition, long-span bridges are flexible structures with crowded modal frequencies. There exist certain number of VIV-prone modes under normal wind velocities. All these VIV-prone modes should be suppressed. To get the best control scheme in balancing the control efficiency and cost, it is more necessary to consider the synergistic effect among the vibration absorbers to optimally determine the total number, spanwise positions as well as the design parameters of each device. To the best knowledge of the authors, this procedure has never been studied before.

This paper intends to shed some lights in this area. In particular, the governing equations of the bridge-VIV system with spatially distributed TMDIs are derived in modal coordinate in Section 2. A global optimization method to optimally design the bridge-DTMDIs system is also proposed in this section. A real bridge with aerodynamic parameters identified through wind tunnel experiment is adopted as a case study in Section 3. The efficiency of the proposed method is validated through comparisons with conventional mode-by-mode approach in Section 4. The robustness of the control schemes against structural and aerodynamic parameters as well as the influence of inerter span length are further discussed in Section 5. Some concluding remarks are drawn in Section 6.

## 2. Governing equations and optimization

### 2.1. Modeling of bridge-DTMDIs system

A bridge equipped with spatially distributed TMDIs is shown in Fig. 2. In the present study, two TMDI installation configurations are considered. In configuration 1, inerter is linked to the same spanwise position with the mass block (Section A). In configuration 2, the inerter spans to a different spanwise position with the mass block (Sections B and C). In the figure,  $f_{wind}(x, t)$  denotes the wind load imposed on the spanwise position  $x$  of the deck.  $y(x, t)$  is the vertical displacement of the

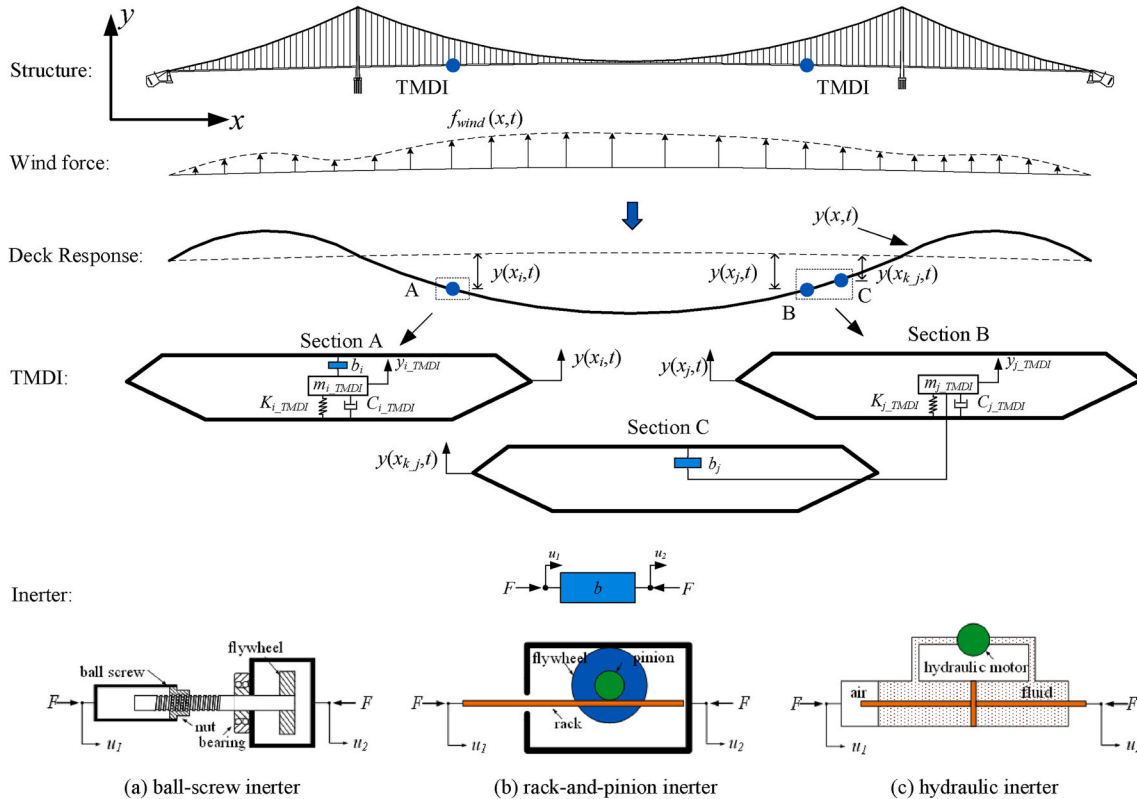


Fig. 2. Arrangement of DTMDIs in long-span bridges.

deck at  $x$ .  $x_j$  is the spanwise position where the  $j$ th TMDI is installed at.  $m_{j\_TMDI}$ ,  $K_{j\_TMDI}$ ,  $C_{j\_TMDI}$ ,  $b_j$  are the mass, stiffness, damping, and inertia of the  $j$ th TMDI, respectively.  $y_{j\_TMDI}(t)$  is the displacement of  $m_{j\_TMDI}$ .  $y(x_j, t)$  is deck displacement at  $x_j$ .  $x_{k-j}$  is the spanwise position where the other terminal of inerter in the  $j$ th TMDI is connected to ( $x_{k-j} = x_j$  if the inerter span length is zero).  $y(x_{k-j}, t)$  is deck displacement at  $x_{k-j}$ .

For any virtual displacements of the deck  $\delta y(x, t)$  and the TMDI  $\delta y_{j\_TMDI}(t)$ , the bridge-DTMDIs system satisfies the virtual work principle, which gives

$$\begin{aligned} & \int_L [f_{wind}(x, t) - m(x)\ddot{y}(x, t) - C(x)\dot{y}(x, t)] \cdot \delta y(x, t) dx \\ & + \sum_j \left[ K_{j\_TMDI} \cdot y_{j\_TMDI}(t) - K_{j\_TMDI} \cdot y(x_j, t) + C_{j\_TMDI} \cdot \dot{y}_{j\_TMDI}(t) \right. \\ & \quad \left. - C_{j\_TMDI} \cdot \dot{y}(x_j, t) + b_j \cdot \ddot{y}_{j\_TMDI}(t) \frac{y(x_{k-j}, t)}{y(x_j, t)} - b_j \cdot \ddot{y}(x_{k-j}, t) \frac{y(x_{k-j}, t)}{y(x_j, t)} \right] \cdot \delta y(x_j, t) \\ & - \sum_j \left[ m_{j\_TMDI} \cdot \ddot{y}_{j\_TMDI}(t) + K_{j\_TMDI} \cdot y_{j\_TMDI}(t) - K_{j\_TMDI} \cdot y(x_j, t) \right. \\ & \quad \left. + C_{j\_TMDI} \cdot \dot{y}_{j\_TMDI}(t) - C_{j\_TMDI} \cdot \dot{y}(x_j, t) + b_j \cdot \ddot{y}_{j\_TMDI}(t) - b_j \cdot \ddot{y}(x_{k-j}, t) \right] \cdot \delta y_{j\_TMDI}(t) \\ & = \int_L EI(x) y''(x, t) \cdot \delta y''(x, t) dx \end{aligned} \quad (1)$$

where  $m(x)$  is the unit mass of the deck at spanwise position  $x$ ;  $L$  the total length of the bridge;  $C(x)$  the damping of the bridge at  $x$ ;  $EI(x)$  the

$$\begin{cases} y(x, t) = \varphi_i(x) \cdot \eta_i(t) \\ y_{j\_TMDI}(t) = 1 \cdot \eta_{j\_TMDI}(t) \\ \delta y(x, t) = \varphi_i(x) \cdot \delta \eta_i(t) \\ \delta y_{j\_TMDI}(t) = 1 \cdot \delta \eta_{j\_TMDI}(t) \\ y''(x, t) = \varphi_i''(x) \cdot \eta_i(t) \end{cases} \quad (2)$$

where  $\varphi_i(x)$  and  $\eta_i(t)$  are modal shape and generalized coordinate of the  $i$ th mode, respectively.

Combining Eq. (2) with Eq. (1), the governing equations for the  $i$ th mode of the bridge with DTMDIs can be obtained:

$$\begin{aligned} & \delta \eta_i(t) \cdot \left[ M_i \cdot \ddot{\eta}_i(t) + C_{ni} \cdot \dot{\eta}_i(t) + K_{ni} \cdot \eta_i(t) \right] \\ & - \delta \eta_i(t) \cdot \left\{ \begin{aligned} & [\varphi_i(x_j)] \cdot [K_{j\_TMDI}] \cdot [\eta_{j\_TMDI}(t)] + [\varphi_i(x_j)] \cdot [C_{j\_TMDI}] \cdot [\dot{\eta}_{j\_TMDI}(t)] \\ & + [\varphi_i(x_{k-j})] \cdot [b_j] \cdot [\dot{\eta}_{j\_TMDI}(t)] - [\varphi_i(x_j)] \cdot [K_{j\_TMDI}] \cdot [\varphi_i(x_j)]^T \cdot \eta_i(t) \\ & - [\varphi_i(x_j)] \cdot [C_{j\_TMDI}] \cdot [\varphi_i(x_j)]^T \cdot \dot{\eta}_i(t) - [\varphi_i(x_{k-j})] \cdot [b_j] \cdot [\varphi_i(x_{k-j})]^T \cdot \ddot{\eta}_i(t) \end{aligned} \right\} \\ & + [\delta \eta_{j\_TMDI}(t)]^T \cdot \left\{ \begin{aligned} & [m_{j\_TMDI}] \cdot [\ddot{\eta}_{j\_TMDI}(t)] + [K_{j\_TMDI}] \cdot [\eta_{j\_TMDI}(t)] \\ & + [C_{j\_TMDI}] \cdot [\dot{\eta}_{j\_TMDI}(t)] + [b_j] \cdot [\ddot{\eta}_{j\_TMDI}(t)] \\ & - [K_{j\_TMDI}] \cdot [\varphi_i(x_j)]^T \cdot \eta_i(t) - [C_{j\_TMDI}] \cdot [\varphi_i(x_j)]^T \cdot \dot{\eta}_i(t) \\ & - [b_j] \cdot [\varphi_i(x_{k-j})]^T \cdot \ddot{\eta}_i(t) \end{aligned} \right\} \\ & = \delta \eta_i(t) \cdot F_i \end{aligned} \quad (3)$$

stiffness of the bridge at  $x$ . The dot symbol denotes time derivative. The apostrophe symbol denotes spatial derivative with respect to  $x$ .

Field measurements indicate that VIV of the bridge is a single mode phenomenon. The above equations can be thus projected to any mode of the bridge. For example, for the  $i$ th mode:

where  $M_i$ ,  $C_{ni}$ ,  $K_{ni}$  and  $F_i$  are the modal mass, modal damping, modal stiffness, and modal force of the  $i$ th mode, respectively. The matrix elements in the above equation are shown in the Appendix.

The above equation can be rewritten as:

$$\begin{aligned}
& \left\{ \begin{matrix} \delta \eta_i(t) \\ [\delta \eta_{j-TMDI}(t)] \end{matrix} \right\}^T \left\{ \begin{matrix} M_i \ddot{\eta}_i(t) - [\varphi_i(x_{k-j})][b_j] \left[ \ddot{\eta}_{j-TMDI}(t) \right] + [\varphi_i(x_{k-j})][b_j][\varphi_i(x_{k-j})]^T \ddot{\eta}_i(t) \\ [m_{j-TMDI}] \left[ \ddot{\eta}_{j-TMDI}(t) \right] + [b_j] \left[ \ddot{\eta}_{j-TMDI}(t) \right] - [b_j][\varphi_i(x_{k-j})]^T \ddot{\eta}_i(t) \end{matrix} \right\} + \\
& \left\{ \begin{matrix} \delta \eta_i(t) \\ [\delta \eta_{j-TMDI}(t)] \end{matrix} \right\}^T \left\{ \begin{matrix} C_{ni} \dot{\eta}_i(t) - [\varphi_i(x_j)][C_{j-TMDI}][\dot{\eta}_{j-TMDI}(t)] + [\varphi_i(x_j)][C_{j-TMDI}][\varphi_i(x_j)]^T \dot{\eta}_i(t) \\ [C_{j-TMDI}][\dot{\eta}_{j-TMDI}(t)] - [C_{j-TMDI}][\varphi_i(x_j)]^T \dot{\eta}_i(t) \end{matrix} \right\} + \\
& \left\{ \begin{matrix} \delta \eta_i(t) \\ [\delta \eta_{j-TMDI}(t)] \end{matrix} \right\}^T \left\{ \begin{matrix} K_{ni} \eta_i(t) - [\varphi_i(x_j)][K_{j-TMDI}][\eta_{j-TMDI}(t)] + [\varphi_i(x_j)][K_{j-TMDI}][\varphi_i(x_j)]^T \eta_i(t) \\ [K_{j-TMDI}][\eta_{j-TMDI}(t)] - [K_{j-TMDI}][\varphi_i(x_j)]^T \eta_i(t) \end{matrix} \right\} = \left\{ \begin{matrix} \delta \eta_i(t) \\ [\delta \eta_{j-TMDI}(t)] \end{matrix} \right\}^T \left\{ \begin{matrix} F_i \\ \mathbf{0} \end{matrix} \right\}
\end{aligned} \quad (4)$$

Defining  $\Delta \eta_{j-TMDI} = \eta_{j-TMDI} - \varphi_i(x_j) \cdot \eta_i$  as the relative displacement of TMDI mass block with respect to the deck, and dividing the vector of virtual displacement on both sides of the equation, it obtains:

the classical TMD (Marian and Giaralis, 2014). That is, if the inertance term in TMDI is null ( $b = 0$ ), TMDI retrieves to the classical TMD. While, if the mass in TMDI is null ( $m = 0$ ), the TMDI retrieves to the TID. Thus,

$$\begin{aligned}
& \left[ \begin{matrix} M_i + [\varphi_i(x_{k-j})][b_j][\varphi_i(x_{k-j})]^T & -[\varphi_i(x_{k-j})][b_j] \\ -[b_j][\varphi_i(x_{k-j})]^T & [m_{j-TMDI}] + [b_j] \end{matrix} \right] \cdot \left[ \begin{matrix} 1 & \mathbf{0} \\ [\varphi_i(x_j)]^T & [1] \end{matrix} \right] \left[ \begin{matrix} \ddot{\eta}_i(t) \\ [\Delta \ddot{\eta}_{j-TMDI}(t)] \end{matrix} \right] \\
& + \left[ \begin{matrix} C_{ni} + [\varphi_i(x_j)][C_{j-TMDI}][\varphi_i(x_j)]^T & -[\varphi_i(x_j)][C_{j-TMDI}] \\ -[C_{j-TMDI}][\varphi_i(x_j)]^T & [C_{j-TMDI}] \end{matrix} \right] \cdot \left[ \begin{matrix} 1 & \mathbf{0} \\ [\varphi_i(x_j)]^T & [1] \end{matrix} \right] \left[ \begin{matrix} \dot{\eta}_i(t) \\ [\Delta \dot{\eta}_{j-TMDI}(t)] \end{matrix} \right] \\
& + \left[ \begin{matrix} K_{ni} + [\varphi_i(x_j)][K_{j-TMDI}][\varphi_i(x_j)]^T & -[\varphi_i(x_j)][K_{j-TMDI}] \\ -[K_{j-TMDI}][\varphi_i(x_j)]^T & [K_{j-TMDI}] \end{matrix} \right] \cdot \left[ \begin{matrix} 1 & \mathbf{0} \\ [\varphi_i(x_j)]^T & [1] \end{matrix} \right] \left[ \begin{matrix} \eta_i(t) \\ [\Delta \eta_{j-TMDI}(t)] \end{matrix} \right] \\
& = \left\{ \begin{matrix} F_i \\ \mathbf{0} \end{matrix} \right\}
\end{aligned} \quad (5)$$

Multiplying the following term on both sides of the above equation:

$$\left[ \begin{matrix} 1 & \mathbf{0} \\ [\varphi_i(x_j)]^T & [1] \end{matrix} \right]^T = \left[ \begin{matrix} 1 & [\varphi_i(x_j)] \\ \mathbf{0} & [1] \end{matrix} \right] \quad (6)$$

The governing equations for the  $i$ th mode of the bridge with DTMDIs can finally be expressed as:

the above governing equations can be treated as a generalized form for line-like structures with multiple TMDIs, TIDs, TMDs, or their combinations.

## 2.2. Modeling of VIV force

The amplitude of VIV displacement and/or acceleration is normally

$$\begin{aligned}
& \left[ \begin{matrix} M_i + [\varphi_i(x_j)][m_{j-TMDI}][\varphi_i(x_j)]^T + [\varphi_i(x_j)(1 - \Delta \varphi_j)][b_j][\varphi_i(x_j)(1 - \Delta \varphi_j)]^T & [\varphi_i(x_j)][m_{j-TMDI}] + [\varphi_i(x_j)(1 - \Delta \varphi_j)][b_j] \\ [m_{j-TMDI}][\varphi_i(x_j)]^T + [b_j][\varphi_i(x_j)(1 - \Delta \varphi_j)]^T & [m_{j-TMDI}] + [b_j] \end{matrix} \right] \\
& \times \left[ \begin{matrix} \ddot{\eta}_i(t) \\ [\Delta \ddot{\eta}_{j-TMDI}(t)] \end{matrix} \right] + \left[ \begin{matrix} C_{ni} & \\ & [C_{j-TMDI}] \end{matrix} \right] \cdot \left[ \begin{matrix} \dot{\eta}_i(t) \\ [\Delta \dot{\eta}_{j-TMDI}(t)] \end{matrix} \right] + \left[ \begin{matrix} K_{ni} & \\ & [K_{j-TMDI}] \end{matrix} \right] \cdot \left[ \begin{matrix} \eta_i(t) \\ [\Delta \eta_{j-TMDI}(t)] \end{matrix} \right] \\
& = \left\{ \begin{matrix} F_i \\ \mathbf{0} \end{matrix} \right\}
\end{aligned} \quad (7)$$

where  $\Delta \varphi_j = \varphi_i(x_{k-j})/\varphi_i(x_j)$  denotes the modal shape ratio between two deck sections where the two terminals of the inerter in  $j$ th TMDI are linked to. It is obvious that if  $\Delta \varphi_j = 0$ , the inerter is linked to the ground or fixed point of a modal shape. While, if  $\Delta \varphi_j = 1$ , the inerter is linked to the same spanwise position with the mass block. As can be seen, the  $j$ th TMDI can make most use of the mass block when installed at the generalized coordinate of the deck ( $\varphi_i(x_j) = 1$ ). Besides, if the inerter spans any length from the generalized coordinate ( $\Delta \varphi_j < 1$ ), the offered virtual mass by the inerter to the mass block will be further enhanced.

It is noticeable that TMDI is often interpreted as a generalization of

adopted as a criterion for assessing the VIV performance of a structure. In the process of DTMDIs optimization, the VIV amplitude of displacement and/or acceleration needs to be repeatedly calculated at different sets of TMDI parameters. Thus, a spectral VIV force model following the framework of Vickery and Basu (1983) is adopted. Comparing with the other forms of VIV force models, the VIV responses of the bridge can be calculated through matrix operations instead of numerical integration, which can significantly improve the computing efficiency.

The VIV force ( $F_i$ ) consists of two terms: the vortex-shedding term and the motion-induced term. The power spectral density (PSD) function of the vortex-shedding term is often defined as (Strommen and

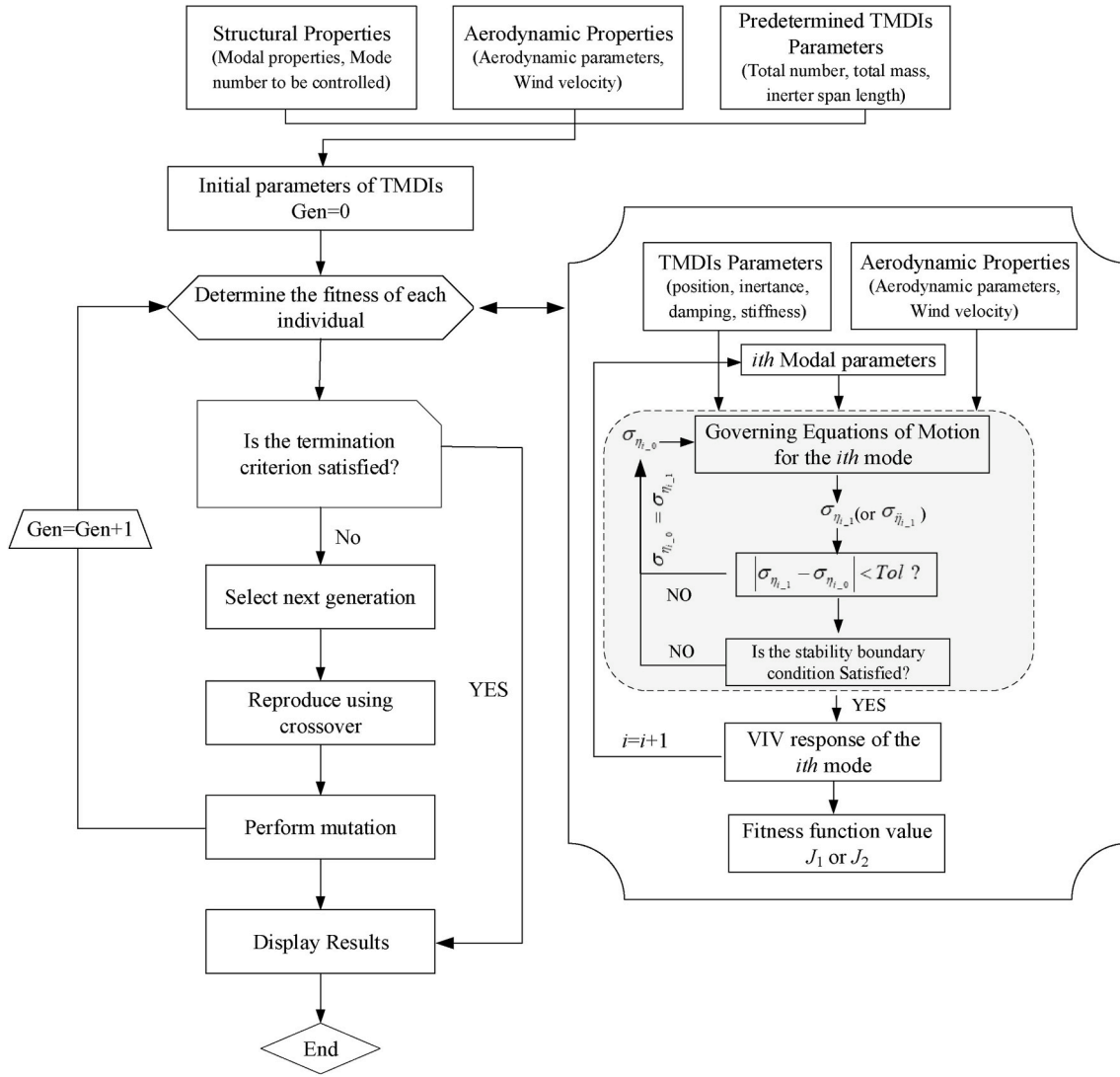


Fig. 3. The flowchart of global optimization.

Hjorth-Hansen, 2001):

$$S_{F_{cl}}(\omega) = \frac{q_v^2 D^2 \sigma_{cl}^2}{\sqrt{\pi} \omega_s B} \exp \left[ - \left( \frac{1 - \omega / \omega_s}{B} \right)^2 \right] \int_0^L \int_0^L R_{cor}(x_1, x_2) \varphi_i(x_1) \varphi_i(x_2) dx_1 dx_2 \quad (8)$$

where  $q_v = \rho U^2 / 2$  is wind pressure;  $U$  wind velocity;  $\rho$  air density;  $D$  deck height;  $R_{cor}(x_1, x_2)$  spatial coherence function;  $\sigma_{cl}$  RMS of the

fluctuating lift coefficient  $\tilde{C}_L(t)$ ;  $\omega_s$  vortex-shedding frequency ( $\omega_s = 2\pi U S_t / D$ , where  $S_t$  is Strouhal number);  $B$  a non-dimensional load spectrum bandwidth parameter, which can roughly be determined by  $B = \sqrt{0.08^2 + 2I_u^2}$  ( $I_u$  longitudinal turbulence intensity) (Pagnini et al., 2020).

For line-like structures with large value of length-to-width ratio, the integral of the spatial coherence function can be simplified into

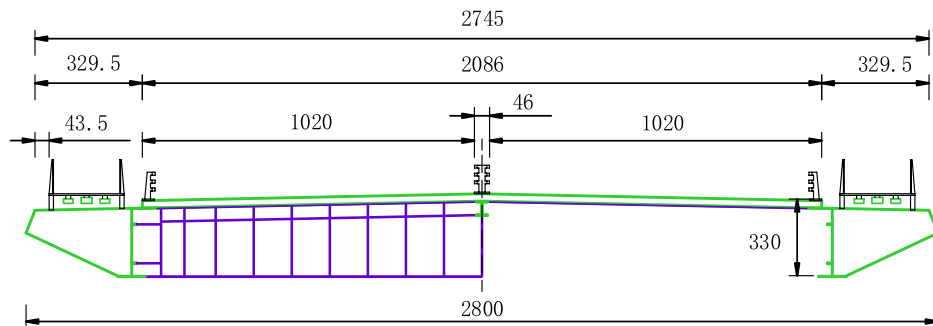


Fig. 4. Cross-section of the deck (unit: cm).



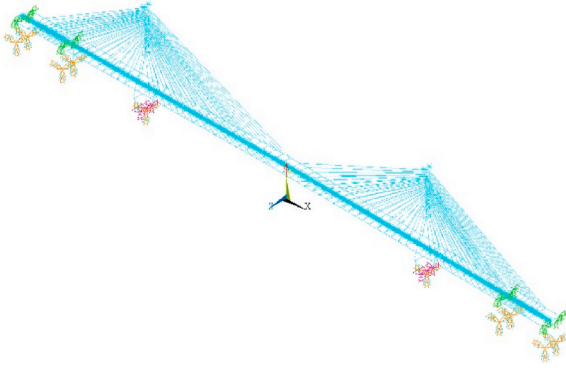


Fig. 5. Finite element model of the bridge.

(Strommen and Hjorth-Hansen, 2001):

$$\int_0^L \int_0^L R_{cor}(x_1, x_2) \varphi_i(x_1) \varphi_i(x_2) dx_1 dx_2 \approx 2\lambda D \int_0^L \varphi_i^2(x) dx \quad (9)$$

where  $\lambda$  is a non-dimensional coherence length scale parameter, which is often found to be near unity or a little bit greater (Dyrbye and Hansen, 1997).

The motion-induced term (the aeroelastic damping) is modeled by:

$$\xi_{ae} = C_{ae} / (2M_i \omega_{ni}) = K_a \frac{\rho D^2}{\bar{m}} \left[ 1 - \left( \frac{\sigma_{\eta_i}}{\gamma D a_L} \right)^2 \right] \quad (10)$$

where  $\bar{m} = M_i / \int_0^L \varphi_i^2(x) dx$  is the equivalent modal mass of the deck per unit length;  $\sigma_{\eta_i}$  root mean square (RMS) of the generalized coordinate  $\eta_i$ ;  $\omega_{ni} = \sqrt{(K_{ni} - K_{ae}) / M_i}$  structural natural frequency;  $K_{ae}$  aeroelastic stiffness, which is often neglected.  $\gamma = \sqrt{\int_0^L \varphi_i(x)^2 dx / \int_0^L \varphi_i(x)^4 dx}$  modal shape correction factor;  $K_a$  linear aeroelastic damping coefficient;  $a_L$  nonlinear aeroelastic damping coefficient.

Combining Eq. (7) with the VIV force model, the governing equations for the  $i$ th mode VIV of the bridge equipped with DTMDIs can be expressed as:

$$\begin{aligned} & \begin{bmatrix} M_i + [\varphi_i(x_j)] [m_{j-TMDI}] [\varphi_i(x_j)]^T + [\varphi_i(x_j)(1 - \Delta\varphi_j)] [b_j] [\varphi_i(x_j)(1 - \Delta\varphi_j)]^T \\ [m_{j-TMDI}] [\varphi_i(x_j)]^T + [b_j] [\varphi_i(x_j)(1 - \Delta\varphi_j)]^T \end{bmatrix} \begin{bmatrix} \varphi_i(x_j) [m_{j-TMDI}] + [\varphi_i(x_j)(1 - \Delta\varphi_j)] [b_j] \\ [m_{j-TMDI}] + [b_j] \end{bmatrix} \\ & \times \begin{bmatrix} \ddot{\eta}_i(t) \\ [\Delta \ddot{\eta}_{j-TMDI}(t)] \end{bmatrix} + \begin{bmatrix} 2M_i \omega_{ni} \xi_{ni} - 2M_i \omega_{ni} \xi_{ae} \\ 2m_{j-TMDI} \omega_{j-TMDI} \xi_{j-TMDI} \end{bmatrix} \cdot \begin{bmatrix} \dot{\eta}_i(t) \\ [\Delta \dot{\eta}_{j-TMDI}(t)] \end{bmatrix} \\ & + \begin{bmatrix} M_i \omega_{ni}^2 \\ [m_{j-TMDI} \omega_{j-TMDI}^2] \end{bmatrix} \cdot \begin{bmatrix} \eta_i(t) \\ [\Delta \eta_{j-TMDI}(t)] \end{bmatrix} = \begin{bmatrix} \tilde{F}_{cli} \\ \mathbf{0} \end{bmatrix} \end{aligned} \quad (11)$$

where  $\xi_{ni} = C_{ni} / (2M_i \omega_{ni})$  is the structural mechanical damping ratio;  $\omega_{j-TMDI} = \sqrt{K_{j-TMDI} / m_{j-TMDI}}$  vibration frequency of the  $j$ th TMDI;  $\xi_{j-TMDI} = C_{j-TMDI} / (2m_{j-TMDI} \omega_{j-TMDI})$  damping ratio of the  $j$ th TMDI.

Then, the mean square displacements or accelerations of the bridge and TMDIs can be obtained:

$$\begin{bmatrix} \sigma_{\eta_i}^2 \\ [\Delta \sigma_{\eta_{j-TMDI}}^2] \end{bmatrix} = \int_{\omega} \text{diag} \left\{ [H(\omega)]^{-1} \begin{bmatrix} S_{F_{cl}}(\omega) & & \\ & \ddots & \\ & & 0 \end{bmatrix} [H(\omega)]^{-1\dagger} \right\} d\omega \quad (12)$$

Table 1

Modal properties of the bridge (first 10th vertical modes).

| Mode number | Modal frequency (rad/s) | Equivalent modal mass of the deck $\bar{m}$ (kg/m) | Modal shape | Position of generalized coordinate (m) <sup>a</sup> |
|-------------|-------------------------|----------------------------------------------------|-------------|-----------------------------------------------------|
| 1st         | 1.935                   | 40899                                              |             | $x_{\eta_1} = 393.2$                                |
| 2nd         | 2.538                   | 44319                                              |             | $x_{\eta_2} = 460.5$                                |
| 3rd         | 3.794                   | 42763                                              |             | $x_{\eta_3} = 391.4$                                |
| 4th         | 4.781                   | 42430                                              |             | $x_{\eta_4} = 438.9$                                |
| 5th         | 5.414                   | 44007                                              |             | $x_{\eta_5} = 120.4$                                |
| 6th         | 5.673                   | 40682                                              |             | $x_{\eta_6} = 669.6$                                |
| 7th         | 6.153                   | 39964                                              |             | $x_{\eta_7} = 247.4$                                |
| 8th         | 6.879                   | 71002                                              |             | $x_{\eta_8} = 757.4$                                |
| 9th         | 6.961                   | 46074                                              |             | $x_{\eta_9} = 31.4$                                 |
| 10th        | 7.760                   | 43456                                              |             | $x_{\eta_{10}} = 758.6$                             |

<sup>a</sup> The left end of the bridge is defined as the origin of the coordinate. The position of generalized coordinate denotes the distance between this point and the left end of the bridge.

$$\begin{bmatrix} \sigma_{\eta_i}^2 \\ [\Delta \sigma_{\eta_{j-TMDI}}^2] \end{bmatrix} = \int_{\omega} \text{diag} \left\{ \omega^4 \cdot [H(\omega)]^{-1} \begin{bmatrix} S_{F_{cl}}(\omega) & & \\ & \ddots & \\ & & 0 \end{bmatrix} [H(\omega)]^{-1\dagger} \right\} d\omega \quad (13)$$

where  $\text{diag}\{\}$  denotes the diagonal of a matrix;  $[H(\omega)]^{-1}$  and  $[H(\omega)]^{-1\dagger}$  are frequency response function of the bridge-DTMDIs system and its complex conjugate, the expression of which is shown in the Appendix.

### 2.3. Global optimization method of the DTMDIs

To get an optimum control efficiency for all VIV-prone modes, two objective functions can be adopted: the sum of non-dimensional RMS displacements and/or the sum of the non-dimensional RMS accelerations of the VIV-prone modes, as defined by:

$$\begin{aligned} J_1 &= \sum_i \frac{\sigma_{\eta_i}}{D} \\ J_2 &= \sum_i \frac{\sigma_{\ddot{\eta}_i}}{D} \end{aligned} \quad (14)$$

The global optimization flowchart is shown in Fig. 3. The total number, total mass, and inerter span length of the DTMDIs are

predetermined. The mass, inertance, spanwise position, frequency as well as damping of each TMDI are to be optimized. For a given set of DTMDIs parameters, the response of each VIV mode can be calculated by using the above governing equations. Then the total response of all the VIV-prone modes can be obtained by Eq. (14). The optimum set of DTMDIs parameters can generate the minimum value of  $J_1$  (or  $J_2$ ). According to Eq. (10), the aeroelastic damping term  $\xi_{ae}$  is a nonlinear function of the displacement RMS of the deck. Thus, iteration is needed in the calculation of  $\sigma_{\eta_i}^2$  and  $\Delta\sigma_{\eta_i-TMDI}^2$  at each given set of DTMDIs parameters. It should be noted that the nonlinear iteration may generate a non-realistic solution which can lead to an instability of the VIV system. Thus, a stability boundary should be added during the calculation. That is, the modal damping of the bridge-DTMDIs system should not be less than zero (Gattulli et al., 2001; Yozo and Masato, 1993). The above procedure can be conducted by using constrained optimization algorithm in MATLAB.

The first author used a similar constrained optimization algorithm in the design of the bridge-TMDI system under VIV (Xu et al., 2019). However, the bridge was assumed to be SDOF and only one TMDI was considered. The governing equations in the above study are totally different from the one established in this study. It is a mode-by-mode approach, which cannot consider the synergistic effect among the DTMDIs when applied to control multi-mode vibration of the bridge. On the contrary, the global optimization method proposed in this section can consider not only the synergistic effect among the DTMDIs, but also the spanwise position and mass distribution of the DTMDIs. The efficiency of the global optimization method will be validated through comparison with the mode-by-mode approach in the following section.

### 3. Example bridge model

#### 3.1. Structural properties

A cable-stayed bridge with a total length of 790 m (the main span length is 420 m) is adopted as a case study. The bridge has an open section girder shape with wind fairings on both the leading and trailing edges. The deck height is 3.3 m. The deck width is 28 m. The cross-section of the deck is shown in Fig. 4.

The software ANSYS is adopted to calculate the modal properties of this bridge. The girder, towers, and piers are modeled by BEAM188 element. The cables are modeled by LINK10 element. The secondary dead load is simulated by MASS21 element and attached to the nodes of the main girder. The sag effect of the cables is considered through the Ernst equation. The bottoms of the towers, transition piers, and side piers are all fixed. The vertical and lateral DOFs of the main girder at the positions of transition and side piers are coupled with the top nodes of these piers. The longitudinal DOF of the main girder is released (floating system). The finite element model of this bridge is shown in Fig. 5. The modal properties of the first 10th vertical modes are shown in Table 1.

#### 3.2. Aerodynamic parameters

Cross-sectional model test on this bridge was conducted in TJ-1 wind tunnel of Tongji University, China. The VIV responses of the cross-sectional model at several Scruton numbers ( $S_c = 2m\xi_n / (\rho D^2)$ ,  $m$  is the mass of the model per unit length,  $\rho$  air density,  $D$  deck height;  $\xi_n$  structural damping ratio) were measured, as shown in Fig. 6. The experiment was reported in a former study by the first author (Xu et al., 2018). Only the results related with aerodynamic parameter identification are presented.

According to Pagnini et al. (2020), the non-dimensional VIV response of the bridge can be expressed by:

$$\frac{\sigma_{\eta_i}}{D} = \sqrt{c_{1F} + \sqrt{c_{1F}^2 + c_{2F}}} \quad (15)$$

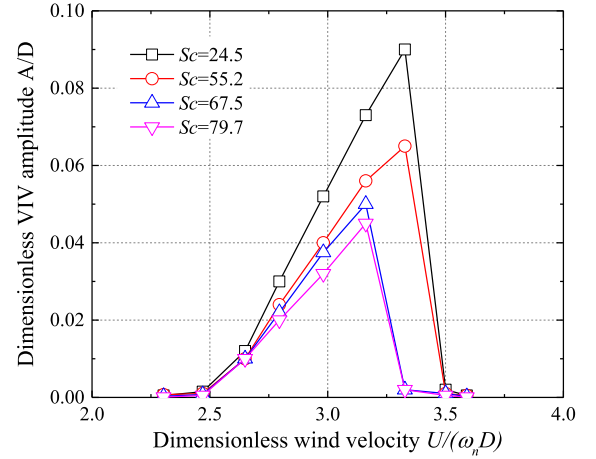


Fig. 6. VIV responses of the sectional model at different Scruton numbers.

where  $c_{1F}$  and  $c_{2F}$  are dimensionless quantities, as defined by:

$$c_{1F} = \gamma^2 \frac{a_L^2}{2} \left( 1 - \frac{S_c}{2K_a} \right) \quad (16)$$

$$c_{2F} = \frac{\sqrt{\pi}}{16^2 \pi^4} \frac{\rho D^2}{m} \frac{a_L^2}{K_a} \gamma^2 C_c^2 \quad (17)$$

$$C_c = \frac{\sigma_{cl}}{\sqrt{BS_t^2}} \cdot coh = \frac{\sigma_{cl}}{\sqrt{BS_t^2}} \cdot \sqrt{\frac{2\lambda D}{\int_0^L \varphi_i^2(x) dx}} \quad (18)$$

For cross-sectional model, the modal shape  $\varphi_i(x) = 1$ . Substituting  $\varphi_i(x) = 1$  into Eq. (16)–Eq. (18), then the values of  $K_a$ ,  $a_L$  and  $C_c$  can be identified through fitting Eq. (15) with experimental results. Based on the VIV property of the structure, there exist two limit boundaries for aerodynamic parameters when using Eq. (15) to fit the experimental results. One is the low-amplitude VIV where the vortex-shedding effect dominates, and the other is the high-amplitude VIV where the motion-induced effect dominates. According to the experimental observations, the cross-sectional model showed typical limit cycle oscillation (LCO). Thus, the high-amplitude VIV boundary is added in the identification of aerodynamic parameters, as defined by (Pagnini et al., 2020):

$$2K_a \left( 1 - \frac{2\sqrt{5}}{a_L^2} \sqrt{c_{2F}} \right) \geq S_c \quad (19)$$

The identified aerodynamic parameters are summarized in Table 2. The simulated VIV responses of the cross-sectional model are compared with the experimental results, as shown in Fig. 7.

The identified vortex-shedding term  $C_c$  consists of two parts:  $\sigma_{cl}/\sqrt{BS_t^2}$  and  $coh$ . The parameter  $B$  determines the bandwidth of the VIV response. As mentioned above, this non-dimensional parameter can be roughly determined by  $B = \sqrt{0.08^2 + 2I_u^2}$  (Pagnini et al., 2020). Thus, a value of 0.08 is chosen. As to the spanwise coherence coefficient  $coh$ , it depends on the coherence length scale parameter  $\lambda$ . Since the spatial coherence experiment on this deck model is lack, a value of  $\lambda = 1$  is

Table 2  
Identified aerodynamic parameters through experiment.

| $U/(D\omega_n)$ | $C_c$    | $K_a$    | $a_L$  |
|-----------------|----------|----------|--------|
| 3.33            | 146.2869 | 43.8875  | 0.0748 |
| 3.16            | 146.2869 | 52.2004  | 0.0589 |
| 2.98            | 146.2869 | 56.0262  | 0.0415 |
| 2.79            | 146.2869 | 70.5819  | 0.0215 |
| 2.65            | 146.2869 | 144.6623 | 0.0082 |



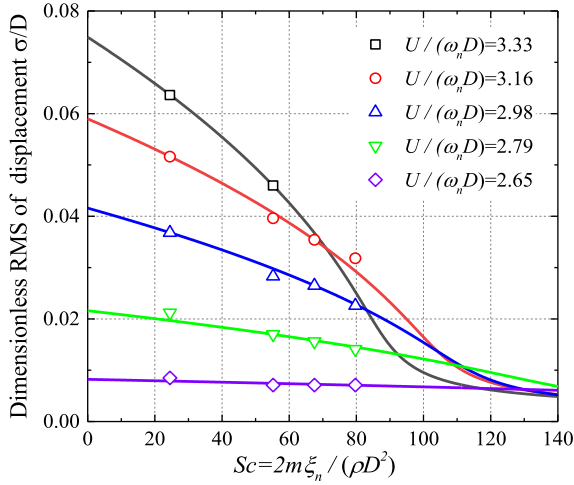


Fig. 7. Comparison between the fitted results and experimental results.

selected as suggested by Dyrbye and Hansen (1997). The deck height of the cross-sectional model is  $D = 0.055$  m, and the length of the cross-sectional model is  $L = 1.74$  m. The calculated spanwise coherence coefficient for the cross-sectional model is  $coh = 0.25$ . After that, the value of  $\sigma_{cl}/S_t^2$  can be determined. It is worth noting that the value of  $\lambda = 1$  was derived for line-like structures with large length-to-width ratios. For cross-sectional models, it may underestimate the spatial coherence of the vortex-shedding term, and thus give a smaller value of  $coh$ . Since the identified value of  $C_c$  is fixed, a smaller value of  $coh$  will lead to an overestimate of the magnitude of the fluctuating lift coefficient ( $\sigma_{cl}$ ). However, it should be noted that the relative magnitude of  $coh$  and  $\sigma_{cl}$  does not influence the values of  $K_a$  and  $a_L$ , since the identified value of  $C_c$  is fixed.

Submitting the aerodynamic parameters into the above governing equations, the VIV responses of the bridge can be calculated. Since the value of  $\lambda$  in the above procedure can influence the magnitude of fluctuating lift coefficient. A sensitivity analysis will be conducted in the following section to investigate the influences of over or under estimate of the vortex-shedding term on the VIV response of the bridge and the control efficiency of the DTMDs.

The non-dimensional RMSs of displacement and acceleration of each mode at the most unfavorable VIV wind velocities are shown in Fig. 8. The displacement RMSs of these modes are around 0.06. The slight difference between each mode is related with the modal Scruton number and modal shape. The acceleration RMSs increase with the increment of mode number. This is because the acceleration is proportional to the

square value of modal frequency. The higher the modal frequency, the larger the acceleration amplitude of this mode.

#### 4. Control schemes and comparison

##### 4.1. Control schemes

The design wind velocity for this bridge is  $U_d = 52$  m/s. According to the experimental result, the first 6 vertical modes are prone to VIV below  $U_d$ . It should be noted that the Chinese Wind-resistant Design Specification for Highway Bridges (JTG/T D60-01-2017) (MOT, 2017) stipulates that the maximum wind velocity considered in the assessment of drivers' safety and comfort is 25 m/s. This is because the bridge is normally closed when the wind velocity exceeds 25 m/s. In addition, the buffeting response of the bridge is more concerned than VIV at ultra-high wind velocities. However, since the main aim of this study is to validate the multi-mode control efficiency of the DTMDs, the involved VIV modes should be as many as possible. Thus, all the 6 VIV-prone modes are considered in this case study.

To compare with conventional tuned mass dampers and mode-by-mode approach, three control schemes are designed. Before optimization, the total mass of the control schemes should be predetermined, since an increase in TMD's mass will normally give a better control performance, but will also lead to a higher cost. For practical design of MTMDs in bridge vibration control, the mass ratio is generally limited to 4% (Domaneschi et al., 2015). Thus, the total mass of the control schemes is taken as 0.5% of the total modal mass of the first 6 modes, i. e.,  $\sum m_{j\_TMDi} / \sum_{i=1}^6 M_i = 0.5\%$ . In this situation, the mass ratio between the control schemes and each bridge modal mass will not exceed 4%. The calculated total mass of the control schemes is  $\sum m_{j\_TMDi} = 212.046$  tons. This value is in the same order of magnitude with the TMDs in real applications. For example, the Trans-Tokyo Bay Crossing Bridge adopts 16 TMDs to suppress the 1st and 2nd bending mode VIVs. The total mass of the TMDs is 160 tons (Fujino and Yoshida, 2002). The Chongqi Bridge adopts 32 TMDs to control the 1st and 2nd bending mode VIVs, the total mass of which is 115.2 tons (Meinhardt et al., 2014). Considering that more modes are involved in this study, though the mass value is a little bit larger than those in the above two bridges, it is still practically reasonable.

The three control schemes are designed as follows:

- (1) Scheme 1: using 6 TMDs to control the 6 VIV-prone modes. Each TMD is placed at the generalized coordinate of the corresponding mode, as shown in Fig. 9. The mass of each TMD is predetermined:  $m_{tmd} = \mu_t \sum_{i=1}^6 M_i / 6$  ( $\mu_t = 0.5\%$  is the mass ratio). The stiffness and damping of each TMD are to be optimized.

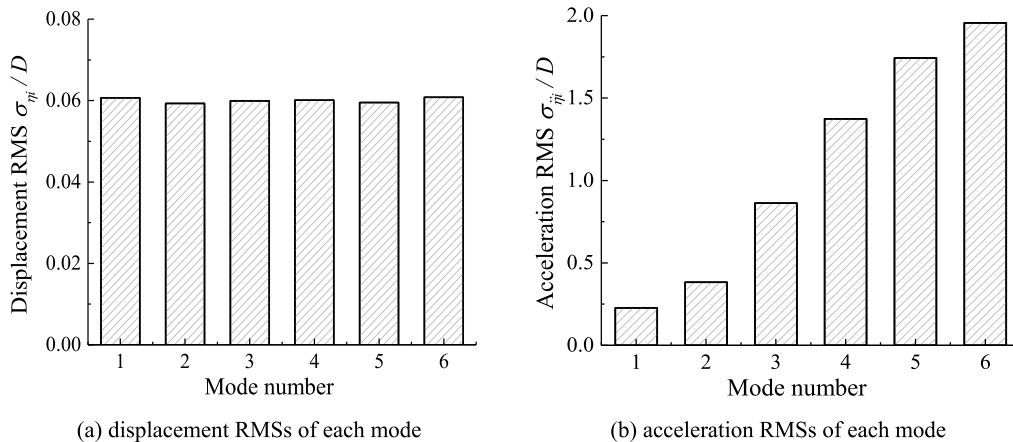


Fig. 8. VIV response of the bridge without control.

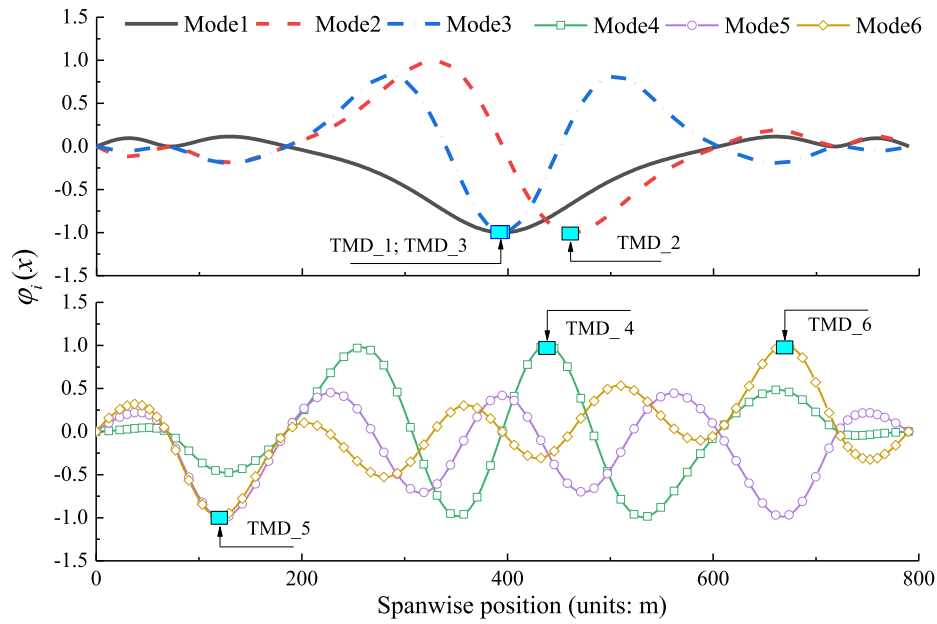


Fig. 9. Positions of the TMDs in Scheme 1.

- (2) Scheme 2: replacing the TMDs in Scheme 1 by TMDIs. The mass of each TMDI is identical with Scheme 1 ( $m_{tmdi} = \mu_t \sum_{i=1}^6 M_i / 6$ ). The stiffness, damping, and inertance of each TMDI are to be optimized.
- (3) Scheme 3: using a smaller number of TMDIs (5 TMDIs) to control the 6 VIV-prone modes. The total mass of the 5 TMDIs equals to the above schemes, but the mass of each TMDI as well as its frequency, damping ratio, inertance, and spanwise position are to be optimized.

According to Fig. 4, the deck height is 3.3 m. To avoid pounding between the mass block and the deck, the static deformation of TMDI (TMD) springs should be restricted. Considering the dimension of mass block and its stroke, an upper limit value of 2 m is selected for the static deformation of TMDI (TMD) springs. Based on this limit value and according to the equation in Fig. 1, the lower limit of vibration frequency for each TMDI (TMD) is 2.21 rad/s. It should be noted that this space limit value is roughly determined based on engineering experience. For a meticulously designed TMD in real engineering project, this limit value

**Table 3**  
Optimized design parameters for each control Scheme.

| Control Scheme |        | Objective function $J_1$ |                       |                                   |                                |                    |                 | Objective function $J_2$ |                       |                                   |                                |                    |                 |
|----------------|--------|--------------------------|-----------------------|-----------------------------------|--------------------------------|--------------------|-----------------|--------------------------|-----------------------|-----------------------------------|--------------------------------|--------------------|-----------------|
|                |        | Mass (kg)                | Spanwise position (m) | Frequency $\omega_{TMDI}$ (rad/s) | Damping ratio $\xi_{TMDI}$ (%) | Inertance $b$ (kg) | $J_1$           | Mass (kg)                | Spanwise position (m) | Frequency $\omega_{TMDI}$ (rad/s) | Damping ratio $\xi_{TMDI}$ (%) | Inertance $b$ (kg) | $J_2$           |
| S1             | TMD_1  | 35341                    | 393.2                 | 2.21 (2.21)                       | 15.98 (15.07)                  | –                  | 0.0584 (0.0596) | 35341                    | 393.2                 | 2.21 (2.21)                       | 15.73 (14.71)                  | –                  | 0.4170 (0.4205) |
|                | TMD_2  | 35341                    | 460.5                 | 2.44 (2.53)                       | 9.67 (3.25)                    | –                  |                 | 35341                    | 460.5                 | 2.47 (2.53)                       | 8.09 (3.25)                    | –                  |                 |
|                | TMD_3  | 35341                    | 391.4                 | 3.78 (3.78)                       | 5.54 (3.52)                    | –                  |                 | 35341                    | 391.4                 | 3.79 (3.79)                       | 3.76 (3.53)                    | –                  |                 |
|                | TMD_4  | 35341                    | 438.9                 | 4.75 (4.77)                       | 3.24 (2.96)                    | –                  |                 | 35341                    | 438.9                 | 4.76 (4.77)                       | 3.06 (2.97)                    | –                  |                 |
|                | TMD_5  | 35341                    | 120.4                 | 5.33 (5.40)                       | 3.16 (3.21)                    | –                  |                 | 35341                    | 120.4                 | 5.34 (5.40)                       | 3.04 (3.22)                    | –                  |                 |
|                | TMD_6  | 35341                    | 669.6                 | 5.74 (5.65)                       | 3.20 (3.65)                    | –                  |                 | 35341                    | 669.6                 | 5.77 (5.66)                       | 3.22 (3.66)                    | –                  |                 |
| S2             | TMDI_1 | 35341                    | 393.2                 | 2.21 (2.21)                       | 3.81 (3.79)                    | 12101 (11181)      | 0.0160 (0.0166) | 35341                    | 393.2                 | 2.22 (2.21)                       | 3.85 (3.80)                    | 12039 (11075)      | 0.2643 (0.2763) |
|                | TMDI_2 | 35341                    | 460.5                 | 2.52 (2.53)                       | 3.30 (3.25)                    | 0                  |                 | 35341                    | 460.5                 | 2.53 (2.53)                       | 3.29 (3.25)                    | 0                  |                 |
|                | TMDI_3 | 35341                    | 391.4                 | 3.78 (3.78)                       | 3.61 (3.52)                    | 0                  |                 | 35341                    | 391.4                 | 3.78 (3.79)                       | 3.54 (3.53)                    | 0                  |                 |
|                | TMDI_4 | 35341                    | 438.9                 | 4.76 (4.77)                       | 3.00 (2.96)                    | 0                  |                 | 35341                    | 438.9                 | 4.76 (4.77)                       | 2.96 (2.97)                    | 0                  |                 |
|                | TMDI_5 | 35341                    | 120.4                 | 5.34 (5.40)                       | 3.16 (3.21)                    | 0                  |                 | 35341                    | 120.4                 | 5.34 (5.40)                       | 3.08 (3.22)                    | 0                  |                 |
|                | TMDI_6 | 35341                    | 669.6                 | 5.74 (5.65)                       | 3.39 (3.65)                    | 0                  |                 | 35341                    | 669.6                 | 5.76 (5.66)                       | 3.29 (3.66)                    | 0                  |                 |
| S3             | TMDI_1 | 41010                    | 394.8                 | 2.22                              | 4.11                           | 14365              | 0.0166          | 21525                    | 393.3                 | 2.23                              | 3.01                           | 7563               | 0.2607          |
|                | TMDI_2 | 36713                    | 460.7                 | 2.53                              | 3.52                           | 0                  |                 | 22970                    | 460.9                 | 2.53                              | 2.88                           | 0                  |                 |
|                | TMDI_3 | 37834                    | 396.5                 | 3.77                              | 3.69                           | 0                  |                 | 30736                    | 395.4                 | 3.77                              | 3.30                           | 0                  |                 |
|                | TMDI_4 | 38093                    | 531.8                 | 4.76                              | 3.23                           | 0                  |                 | 46584                    | 531.8                 | 4.75                              | 3.33                           | 0                  |                 |
|                | TMDI_5 | 58396                    | 669.5                 | 5.51                              | 5.07                           | 0                  |                 | 90231                    | 669.6                 | 5.52                              | 5.72                           | 0                  |                 |

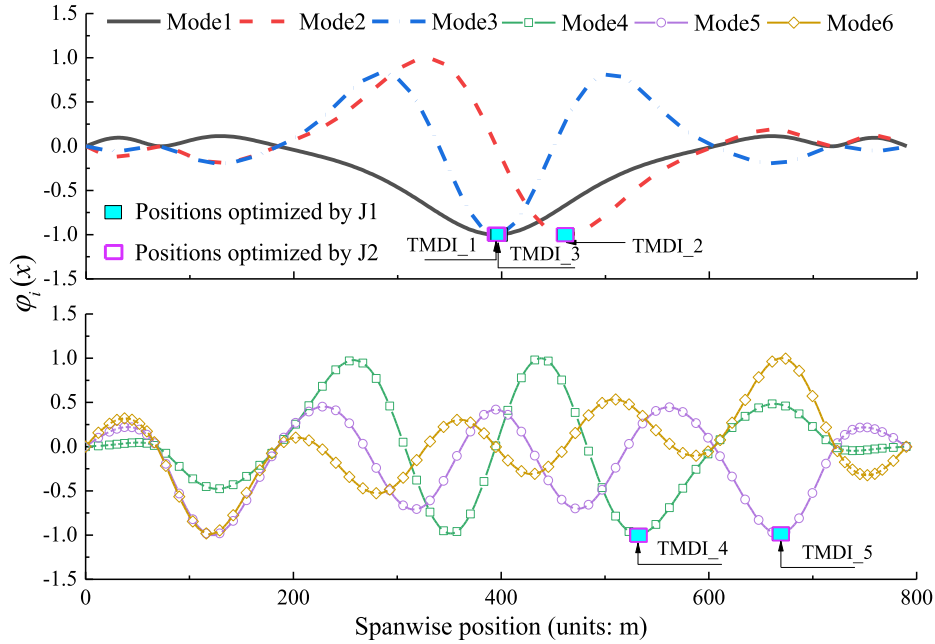


Fig. 10. Optimum positions of the TMDs in Scheme 3.

may be different.

#### 4.2. Optimized design parameters

The optimized parameters of these control schemes are listed in Table 3. For Scheme 1, the TMDs are placed at the positions of generalized coordinates of the bridge modes. The optimized frequencies of these TMDs are close to the corresponding modal frequencies except for TMD\_1. This TMD is restricted by the frequency limit value of 2.21 rad/s to ensure space feasibility. Due to the detuning of TMD\_1, the optimized frequency of TMD\_2 deviates from the natural frequency of Mode 2, and slightly approaches to that of Mode 1 to compensate this influence. In addition, the optimized damping ratios of TMD\_1 and TMD\_2 are obviously larger than the other TMDs. The optimized frequencies of the other TMDs are not influenced by the detuning of TMD\_1. This is due to the narrow-banded property of TMD that the working bandwidth of higher-mode TMDs exceeds the natural frequency of Mode 1. It is noticeable that the optimized frequency of TMD\_5 is slightly smaller than the

natural frequency of Mode 5, and the frequency of TMD\_6 is slightly larger than that of Mode 6. This is because the natural frequencies of these two modes are quite close. To avoid overconcentration of effective frequency in the working bandwidth of the control schemes, the optimization procedure enlarges the frequency difference between these two TMDs.

Scheme 2 replaces the TMDs in Scheme 1 by TMDIs. The optimized frequency of TMDI\_1 also equals to the limit value of 2.21 rad/s. But the frequency of TMDI\_2 no longer moves towards Mode 1. In addition, the optimized damping ratios of TMDI\_1 and TMDI\_2 are close to the other TMDIs. This is because the involving of the inerter solves the detuning problem of TMD in Scheme 1. As will be illustrated later, the control efficiency can be obviously improved. However, it is noticeable that the optimized inertances of TMDIs 2–6 are null, indicating that these TMDIs degrade to conventional TMDs during the optimization. This is because, as indicated by former studies of the first author (Xu et al., 2019, 2020), the space superiority of TMDI over TMD is based on a slight sacrifice of control efficiency. When tuning to Modes 2–6, the TMDIs are not

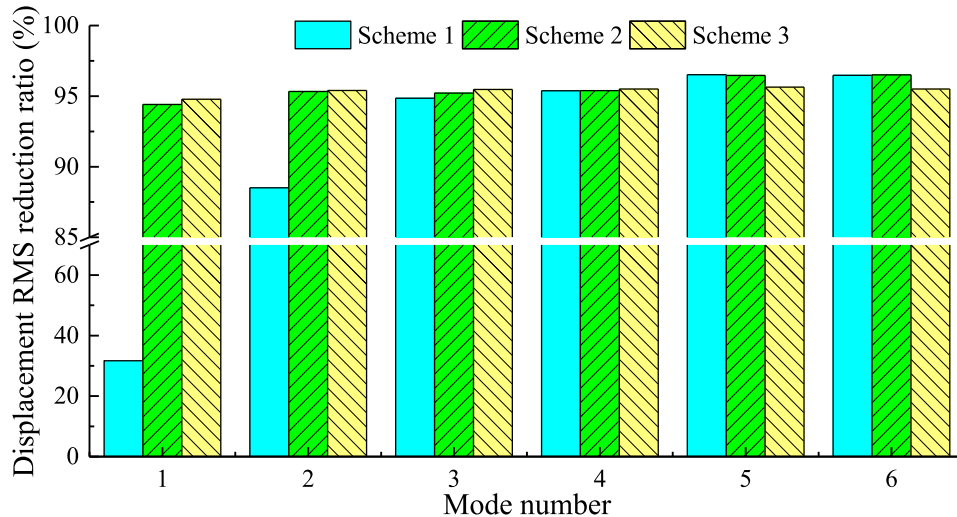


Fig. 11. Displacement RMS reduction ratios by using different control schemes.

restricted by the frequency limit bound. Thus, these TMDIs automatically degrade to TMDs during the optimization.

Scheme 3 seeks to use a smaller number of TMDIs to suppress all the VIV-prone modes. The spanwise positions of these TMDIs are automatically determined during optimization, as shown in Fig. 10. The first three TMDIs are placed at the generalized coordinates of the first three modes and tuned to the corresponding mode. TMDI\_4 moves to at another spanwise position of generalized coordinate and is tuned to Mode 4. And TMDI\_5 is placed to a symmetric position of the generalized coordinate of Mode 5, where the modal shape value of Mode 6 is also quite large. The optimized frequency of this TMDI is 5.51 rad/s, which is between the modal frequencies of the 5th and 6th modes. In this sense, TMDI\_5 is adopted to suppress both the 5th and 6th modes. Since a single TMDI is assigned to control these two modes, the optimized mass of TMDI\_5 is thus the largest among these TMDIs.

The optimized parameters by using  $J_2$  are basically consistent with those obtained by using  $J_1$ . The only difference is the mass distribution in Scheme 3, where the mass of the TMDIs increases with the increment of mode number. As shown in Fig. 8(b), this is because the acceleration amplitude increases with mode number. To get the minimum value of  $J_2$ , more mass is assigned to the TMDIs corresponding to the higher modes.

The designed parameters of Scheme 1 and Scheme 2 by mode-by-mode approach are also shown in the bracket in Table 3. Comparing with the global optimization approach, the TMDs (or TMDIs) are all

tuned to the corresponding modal frequencies, indicating that there is no frequency adjustment among these devices. As discussed above, this is because the mode-by-mode approach cannot consider the synergistic effect among the multiple TMDs (or TMDIs). The values of  $J_1$  and  $J_2$  obtained by mode-by-mode approach are thus larger than those by the global optimization approach, indicating that the control efficiency is worse.

#### 4.3. Control efficiencies

The displacement RMS reduction ratios for each mode, defined as  $(1 - \sigma_{\eta_i, \text{control}} / \sigma_{\eta_i, \text{uncontrol}}) \%$ , are shown in Fig. 11. As shown, the control efficiency of Scheme 1 is the worst among these schemes, especially for Mode 1, the displacement RMS reduction ratio of which is lower than 35%. The worst control efficiency of Mode 1 is due to the detuning of TMD\_1. Besides, since the optimized frequency of TMD\_2 moves towards Mode 1, the displacement reduction ratio of Mode 2 (around 88%) is also smaller than those of the higher modes (around 95%).

The optimized Scheme 2 suggests using one TMDI to control the first mode and five TMDs to control the other modes. The involving of inerter solves the detuning problem of TMD\_1 in Scheme 1, and releases TMD\_2 from deviation. The displacement reduction ratios of the 6 modes are all around 95%. The results indicate that the proposed method is quite efficient for VIV control of long-span bridges. It can optimally determine

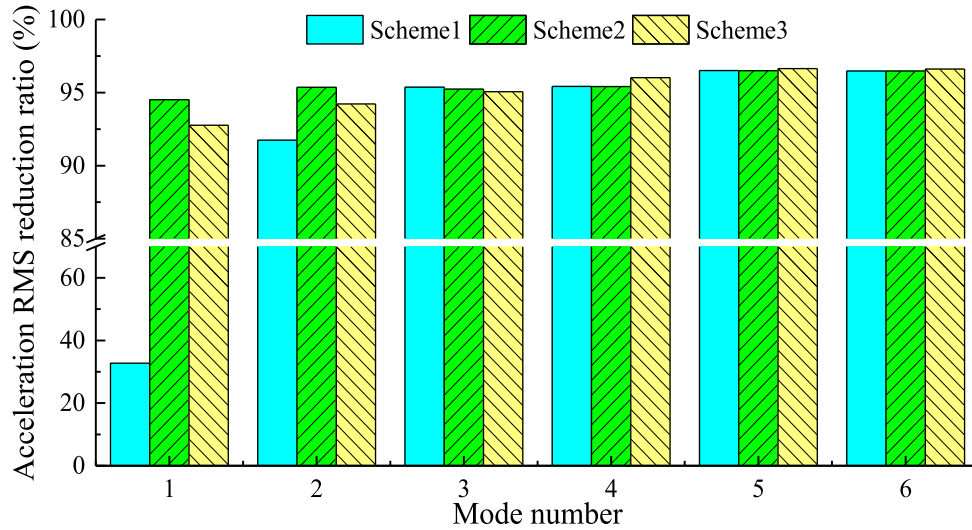


Fig. 12. Acceleration RMS reduction ratios by using different control schemes.

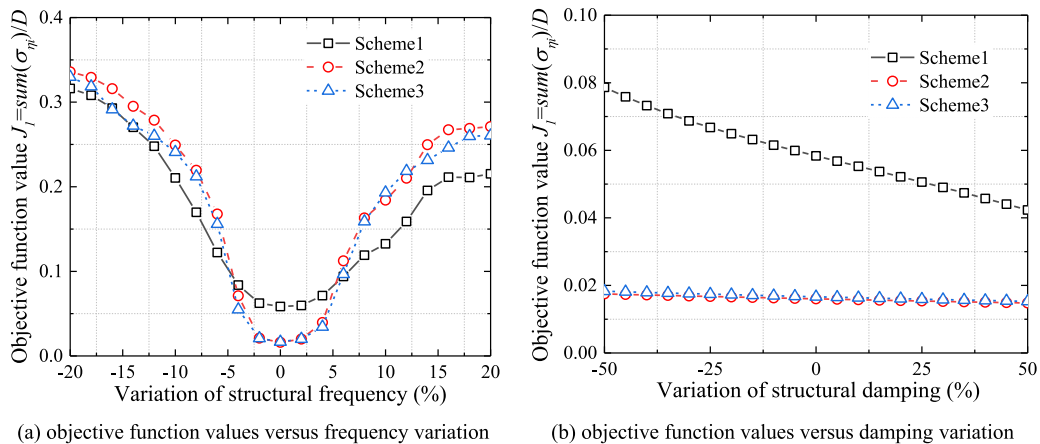


Fig. 13. Sensitivity of the control devices against structural parameters.

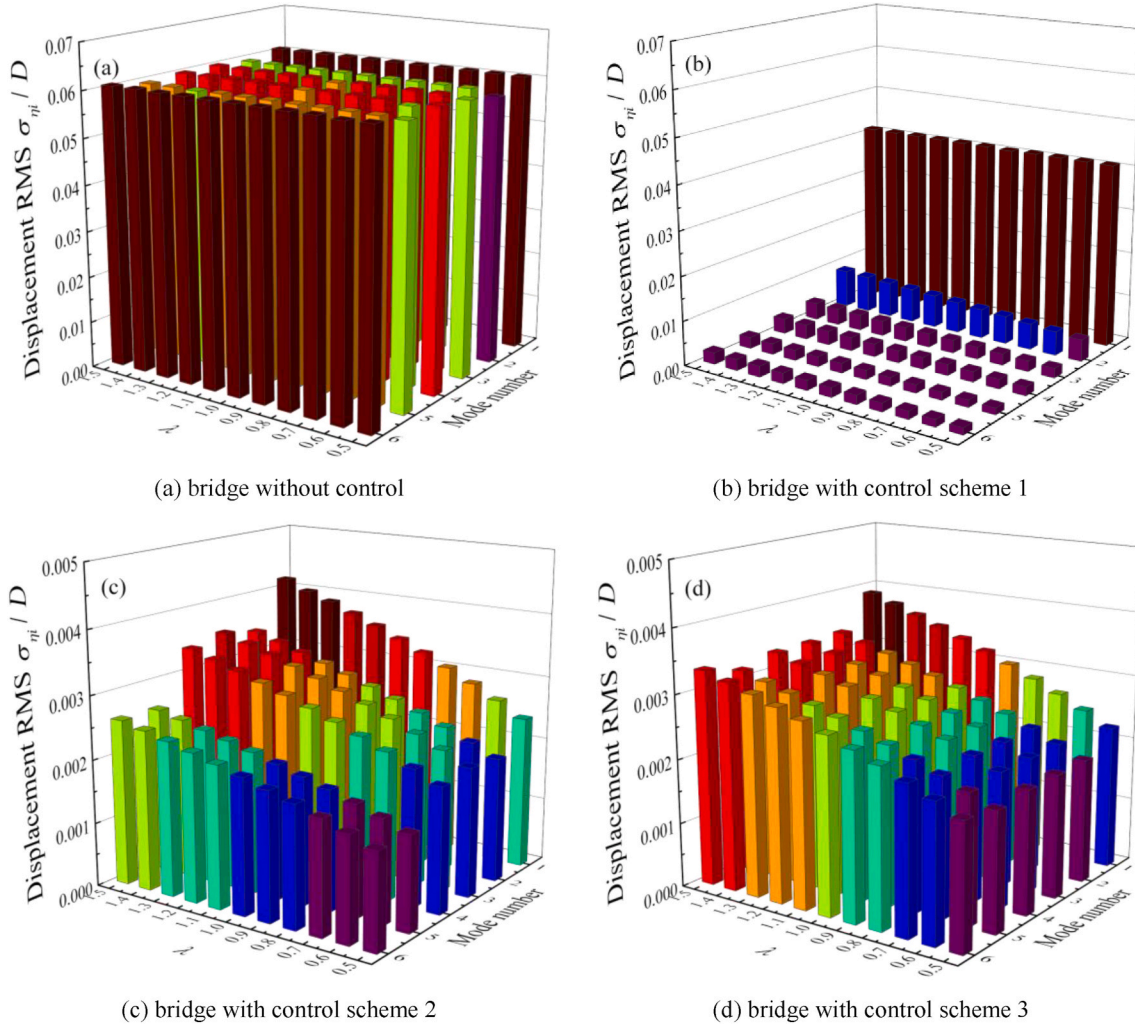


Fig. 14. Sensitivity of the control devices against vortex-shedding term in VIV force model.

the design parameters of the TMDs when the internal space of the deck is restricted, and can also automatically degrade to conventionally TMDs once the internal space of the deck is not restricted.

Scheme 3 seeks to use 5 TMDs to suppress all the 6 modes. The displacement reduction ratios of the first 4 modes are comparable with (or even larger than) Scheme 2. However, since only one TMDI is assigned to control both the 5th and 6th modes, the displacement reduction ratios for these two modes are slightly smaller than those in Scheme 2. Thereby, the objective function value ( $J_1$ ) of Scheme 3 is a little bit larger than that of Scheme 2 (as shown in Table 3).

The acceleration RMS reduction ratios of each mode, defined as  $(1 - \sigma_{\eta_i\_control} / \sigma_{\eta_i\_uncontrol}) \%$ , are shown in Fig. 12. The basic trends for Scheme 1 and Scheme 2 are similar with those designed by  $J_1$ . The only difference is Scheme 3, where the reduction ratios of the 4th to 6th modes increased but the reduction ratios of the 1st to 3rd modes decreased. As shown in Table 3, this is because the mass distribution of TMDIs in Scheme 3 increases with the increment of mode number. Since the acceleration amplitude dramatically increases with mode number, a slight increase in the acceleration reduction ratio for the higher mode will lead to an obvious reduction in the objective function value. This is the reason why the value of  $J_2$  for Scheme 3 is the smallest among these control schemes (as shown in Table 3).

The above results indicate that the proposed global design method can achieve a better control efficiency with even less dampers as compared with conventional mode-by-mode approach for the same mass ratio. However, it should be noted that more design parameters are

involved in the global optimization procedure compared to mode-by-mode approach. Thus, more computational effort is needed.

## 5. Further discussions

### 5.1. Influence of structural parameters

The TMDs/TMDIs are optimized based on the expected values of dynamic properties of the bridge. However, some uncertainties may exist in estimating the properties of the prototype bridge. Moreover, the long-term degradation of the structure will also lead to a variation in the structural properties, especially the modal damping. The robustness against variation of structural properties is an important capacity for control devices. Thus, the sensitivity of these control schemes against structural parameters is investigated in this section.

Fig. 13(a) shows the objective function values  $J_1$  (sum of the non-dimensional displacement RMSs of the 6 VIV modes) at different variations of structural frequency. As can be seen, within a limited frequency variation range ( $\pm 5\%$ ), the control efficiencies of these schemes are insensitive to the structural frequency. And the control efficiencies of Scheme 2 and Scheme 3 are better than that of Scheme 1. While, if the variation of structural frequency exceeds this range, the control efficiencies of these schemes become worse.

If we further compare these schemes at large values of frequency variation, Scheme 1 becomes better than Scheme 2 and Scheme 3. This is because the TMDIs in Scheme 2 and Scheme 3 are all tuned to the



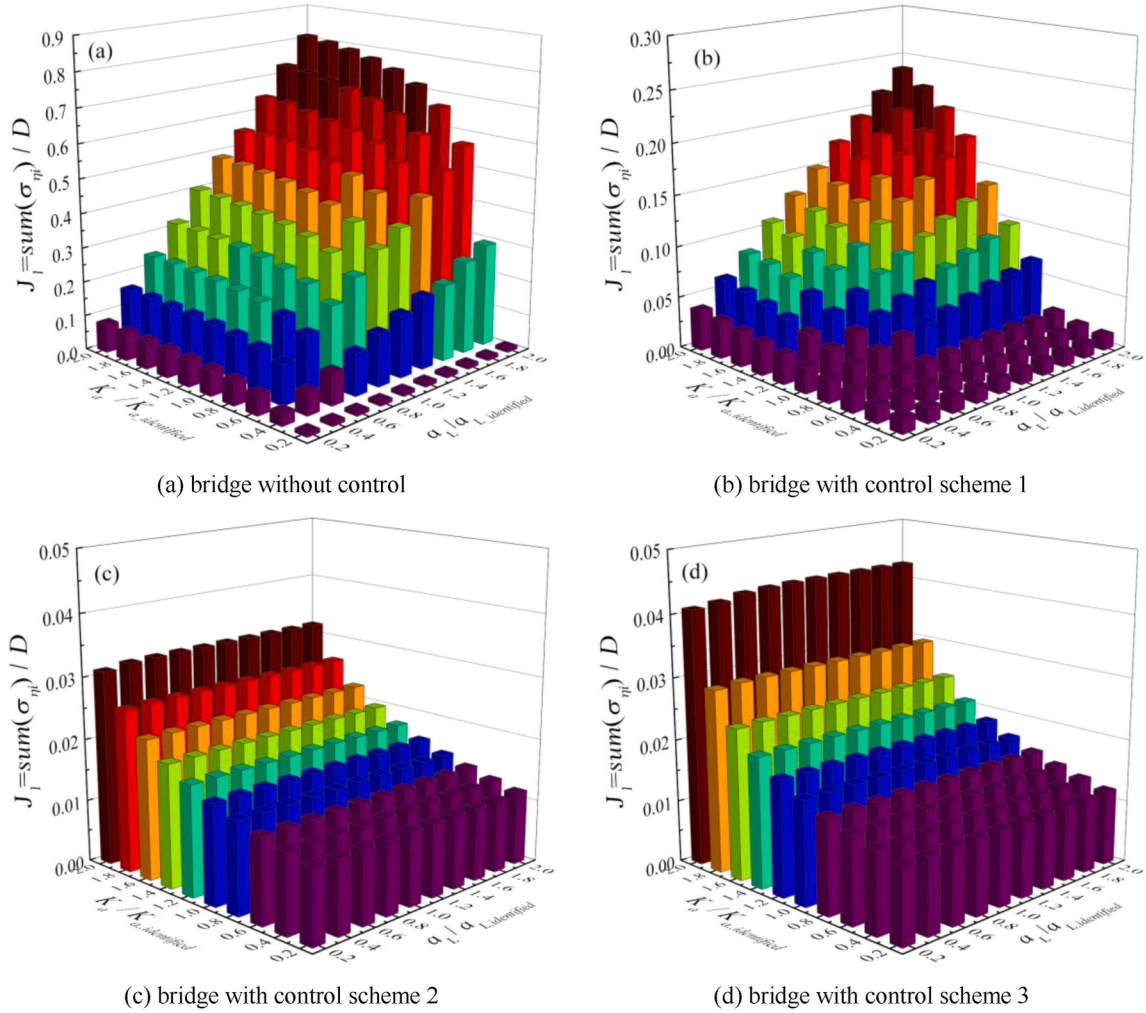


Fig. 15. Sensitivity of the control devices against aerodynamic damping terms in VIV force model.

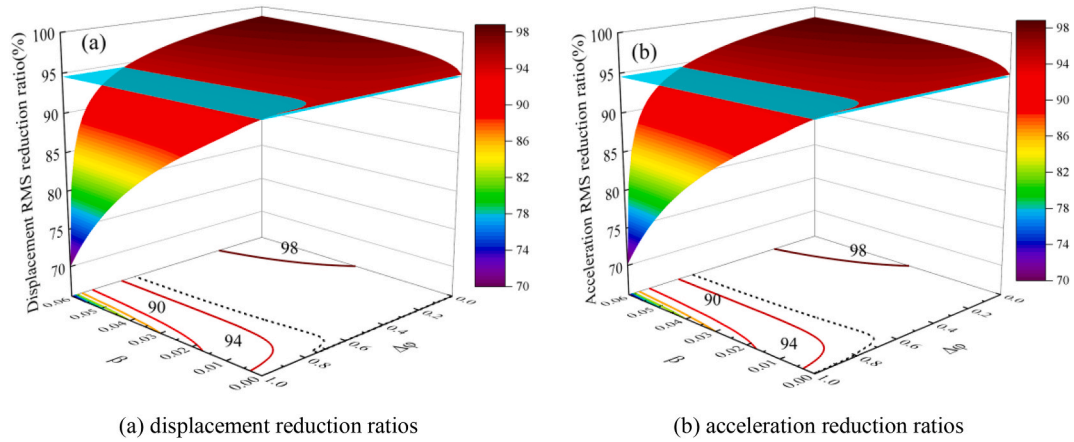


Fig. 16. Control efficiencies of TMDI at different inertia ratios and inerter span lengths.

optimum frequencies to get the best reduction ratio. As a result, the control efficiencies of all the 6 modes will simultaneously decrease with the variation of structural frequency. While for Scheme 1, the frequency of TMD\_1 is detuning, and the frequency of TMD\_2 (2.44 rad/s) slightly approaches to that of Mode 1 to compensate this influence. When the structural frequency decreases, Mode 2 is relatively less influenced compared to the other modes. While, when the structural frequency

increases, the natural frequency of Mode 1 (1.935 rad/s) approaches to the frequency of TMD\_1 (2.21 rad/s) and the control efficiency of this mode becomes better.

Fig. 13(b) shows the influence of structural damping. The objective function values monotonously decrease with the increment of structural damping. This is because the VIV of a structure is governed by the structural mass-damping parameter. And the response magnitude will

decrease with the increment of structural damping. The control efficiencies of Scheme 2 and Scheme 3 are better than Scheme 1 at all variations of damping ratios.

As can be seen, the control efficiencies of these schemes are sensitive to the variation of structural frequency and less sensitive to the variation of structural damping. The globally optimized DTMDs can get a better control efficiency than DTMDs, but are also more sensitive to the variation of structural frequency.

### 5.2. Influence of aerodynamic parameters

In addition to the structural parameters, the VIV response of the bridge is also influenced by the aerodynamic parameters of VIV force models. There also exist some factors that may influence the identification of aerodynamic parameters, such as the accuracy of test and measurement facilities, the identification algorithm (Marra et al., 2011), etc. Besides, the aerodynamic parameters vary with wind velocities within the lock-in region. Thus, the sensitivity of the control schemes against the aerodynamic parameters are also investigated.

Fig. 14 shows the sensitivity of the control scheme against the vortex-shedding term. The magnitude of vortex-shedding term is adjusted through increasing and decreasing the spatial coherence parameter  $\lambda$  in Eq. (9). During the adjustment, the other parameters remain unchanged. Fig. 14(a) shows the displacement RMSs of each mode without control devices. The response of the bridge is insensitive to the variation of vortex-shedding term. This is because the VIV of the bridge is characterized by the high-amplitude VIV, where the motion-induced effect dominates. And the contribution of vortex-shedding term in the total VIV force is quite small. Fig. 14(b) shows the results of the bridge with Scheme 1. Since TMD\_1 in Scheme 1 is detuning, the VIV of Mode 1 is not fully suppressed and is characterized by the high-amplitude VIV. Thus, it is insensitive to the variation of vortex-shedding term. As to the other modes, the VIV cannot fully develop under the action of control devices. Their response is thus characterized by the low-amplitude VIV and monotonously increases with the increment of vortex-shedding term. Fig. 14(c) and (d) show the results of Scheme 2 and Scheme 3, where the VIV of all the 6 modes can be suppressed under the action of control devices. The response of all these modes monotonously increases with the increment of vortex-shedding term.

As can be seen, the variation of vortex-shedding term has little influence on the calculated bridge response if the control device fails to suppress the VIV. While, if the VIV of the bridge can be suppressed by the control device, the variation of vortex-shedding term can influence the calculated magnitude of bridge response. Thus, an over or under estimate of the vortex-shedding term will influence the assessment of the actual displacement (or acceleration) reduction ratios of the control devices. However, it is worth noting that the numerical optimization of the DTMDs is to find the minimum value of the fitness function ( $J_1$  or  $J_2$ ). It relies on the relative magnitudes of these fitness function values at different sets of parameters. Considering that the variation of vortex-shedding term simultaneously influences the calculated response of each mode in Scheme 2 and Scheme 3, its influence on the optimized parameters of the DTMDs would be quite limited. While, for Scheme 1, the variation of vortex-shedding term has little influence on Mode 1, but has some influences on the other modes. The variation of vortex-shedding term may influence the contribution ratio of each mode to the value of fitness function. In this situation, it may have some influence on the optimized parameters of the TMDs. For example, it may influence the deviation of TMD\_2 to Mode 1. However, since this scheme is detuning and is not the optimum control scheme, it will not be further investigated here.

Fig. 15 shows the sensitivity of the control devices against the aeroelastic damping coefficients.  $K_{a\_identified}$  and  $a_{L\_identified}$  denote the linear and nonlinear aeroelastic damping terms identified through test. Fig. 15(a) shows the objective function values  $J_1$  without control devices. The VIV amplitude of the bridge is more sensitive to the nonlinear

aeroelastic damping term and less sensitive to the linear one. According to Eq. (10), this is because the total aeroelastic damping depends on the square value of  $a_L$ . The influence of this term is thus more obvious. Fig. 15(b) shows the results of the bridge with Scheme 1. Since VIV of Mode 1 is not fully suppressed, the influence of aeroelastic damping parameters has a similar trend with the uncontrolled condition. Fig. 15(c) and (d) show the results of Scheme 2 and Scheme 3, where the VIV of all the 6 modes can be suppressed under the action of control devices. Since the VIV of the bridge cannot fully develop, the contribution of the nonlinear aeroelastic damping term to the total aeroelastic damping is quite small. In this situation, the VIV of the bridge becomes more sensitive to the linear aeroelastic damping term, the magnitude of which monotonously increases with the increment of  $K_a$ .

As can be seen, for the bridge-DTMDs system, the calculated response of the bridge is sensitive to the linear aeroelastic damping term and is less sensitive to the nonlinear one. In another word, an over or under estimate of the linear aeroelastic damping term will influence the assessment of the actual displacement (or acceleration) reduction ratios of the control devices. However, it is worth noting again that the 6 modes are simultaneously influenced by the aeroelastic damping term. Thus, its influence on the optimized parameters of the DTMDs would also be quite limited.

### 5.3. Influence of inerter span

In the above section, the inerter is linked to the same spanwise position with the mass block ( $\Delta\varphi_j = 1$ ). According to Eq. (7), if the inerter spans to another spanwise position from the generalized coordinate ( $\Delta\varphi_j < 1$ ), the virtual mass of the inerter imposed to the mass block can be further increased. This trend has been validated in high-rise buildings (Giaralis and Petrini, 2017) and short-span bridges (Dai et al., 2019). Here, in this section, the practical feasibility of inerter span in long-span bridges is further discussed.

For simplicity, only Mode 1 is adopted as a case study. A single TMDI is adopted to control this mode. The design parameters of the TMDI are optimized at different values of inertance ratio  $\beta$  (defined as  $\beta = b/M_j$ ) and span length parameter  $\Delta\varphi$ . As illustrated in Section 2.1,  $\Delta\varphi = 1$  corresponds to the condition where the inerter spans zero length. While  $\Delta\varphi = 0$  corresponds to the condition where the inerter is connected to the ground or fixed-point of the modal shape.

The displacement and acceleration RMS reduction ratios of Mode 1 are shown in Fig. 16. The horizontal plate denotes the reduction ratio by using TMD ( $\beta = 0$ ). As can be seen, there exists a turning line (its projection on the x-y axes is dashed) along which the contribution of inerter span length can be divided into two regions. When the inerter spans a short distance ( $\Delta\varphi > 0.65$ ), the control efficiency of TMDI is inferior to TMD, and is decreased with the increment of inertance ratio. On the other hand, when the inerter spans a large distance ( $\Delta\varphi < 0.65$ ), the control efficiency of TMDI becomes superior to TMD, and is increased with the increment of inertance ratio. In addition, if the inertance ratio  $\beta$  is fixed, the control efficiency of TMDI monotonically increases with the increment of inerter span length.

The above results indicate that the increase of inerter span length can further improve the control efficiency of TMDI. However, to be superior to TMD with the same mass ratio, the span length needs to exceed a critical value. For this case study, the value of  $\Delta\varphi$  should be less than 0.65, indicating that the other terminal of the inerter should be linked to a spanwise position, the modal shape function value of which is less than 0.65 of the generalized coordinate. According to Fig. 9, to achieve this value, the inerter span length should be larger than 70 m.

As can be seen, for long-span bridges, it is practically unfeasible to make the inerter span a long distance to make the control efficiency of TMDI better than TMD. However, for high buildings and short-span bridges (such as girder bridge), the required critical inerter span length is much smaller than that of long-span bridges. In this situation,

the inerter span layout becomes more practically feasible, for example through using the lever-structure suggested by Dai et al. (2019). For these structures, the influence of inerter span length can also be embedded in the proposed governing equations in this study to optimally design the parameters of DTMDIs.

## 6. Conclusions

This paper proposes to use DTMDIs to control multi-mode VIV of long-span bridges. The governing equations of the bridge-DTMDIs system are established in modal coordinate. A global optimization method is proposed to optimally design the parameters of these DTMDIs. The efficiency of the proposed method over conventional mode-by-mode approach is validated through a numerical case study. The robustness of the control schemes against structural and aerodynamic parameters, and the influence of inerter span length are also discussed. The main findings and conclusions are as follows:

- (1) Due to the space restriction, conventional DTMDs can hardly suppress low mode VIV of long-span bridges. Through replacing TMDs by TMDIs, all VIV-prone modes can be suppressed. The global optimization method can consider the synergistic effect among the TMDIs. Comparing with conventional mode-by-mode approach, this method can achieve a better control efficiency with even less dampers.
- (2) The control efficiencies of the control schemes are sensitive to the variation of structural frequency and less sensitive to the variation of structural damping. The globally optimized DTMDIs can get a better control efficiency than DTMDs, but are also more sensitive to the variation of structural frequency.
- (3) The bridge-DTMDIs system is characterized by the low-amplitude VIV, the response of which is sensitive to the magnitudes of vortex-shedding term and linear aeroelastic damping term, and is less sensitive to the nonlinear aeroelastic damping term. An over or under estimate of the aerodynamic parameters will influence the prediction of the actual reduction ratios of the control

devices. However, since the variation of aerodynamic parameters simultaneously influences the calculated response of each mode, its influence on the optimized parameters of the DTMDIs would be quite limited.

- (4) Through spanning the inerter to a different spanwise position, the control efficiency of TMDI can be further improved. But it needs to exceed a critical span length to be superior to TMD with the same mass ratio. For long-span bridges, this layout should be comprehensively judged.
- (5) This study only focuses on suppressing the vertical vibration of the bridge deck, where the TMDIs are arranged at the middle of the deck's cross-section. However, for suppressing torsional vibration or coupled vertical-torsional vibration of the bridge deck, TMDIs symmetrically arranged at the two sides of the deck's cross-section should be adopted. The suppression of this kind of vibrations by using DTMDIs will be conducted in future studies.

## CRedit authorship contribution statement

**Kun Xu:** Methodology, Formal analysis, Experiment, Writing. **Qian Dai:** Calculation, Investigation, Writing – review & editing. **Kaiming Bi:** Validation, Writing – review & editing. **Genshen Fang:** Wind tunnel, Formal analysis, Validation. **Lin Zhao:** Resources, Supervision, Project administration.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix

The matrix elements in Eq. (3) have the following forms:

$$[\boldsymbol{\varphi}_i(x_j)] = [\varphi_i(x_1), \varphi_i(x_2), \dots, \varphi_i(x_{Nd})] \quad (A1)$$

$$[\boldsymbol{\varphi}_i(x_{k-j})] = [\varphi_i(x_{k1-1}), \varphi_i(x_{k2-2}), \dots, \varphi_i(x_{kNd-Nd})] \quad (A2)$$

$$[\boldsymbol{\eta}_{j\_TMDI}(t)] = \begin{bmatrix} \eta_{1\_TMDI}(t) \\ \eta_{2\_TMDI}(t) \\ \vdots \\ \eta_{Nd\_TMDI}(t) \end{bmatrix} \quad (A3)$$

$$[\mathbf{m}_{j\_TMDI}] = \begin{bmatrix} m_{1\_TMDI} & & & \\ & m_{2\_TMDI} & & \\ & & \ddots & \\ & & & m_{Nd\_TMDI} \end{bmatrix} \quad (A4)$$

$$[\mathbf{K}_{j\_TMDI}] = \begin{bmatrix} K_{1\_TMDI} & & & \\ & K_{2\_TMDI} & & \\ & & \ddots & \\ & & & K_{Nd\_TMDI} \end{bmatrix} \quad (A5)$$

$$[\mathbf{C}_{j\_TMDI}] = \begin{bmatrix} C_{1\_TMDI} & & & \\ & C_{2\_TMDI} & & \\ & & \ddots & \\ & & & C_{Nd\_TMDI} \end{bmatrix} \quad (A6)$$

$$[b_j] = \begin{bmatrix} b_1 & & & \\ & b_2 & & \\ & & \ddots & \\ & & & b_{N_d} \end{bmatrix} \quad (A7)$$

where  $N_d$  is the total number of DTMDIs.

The frequency response function of the bridge-DTMDIs system in Eq. (12) has the following form:

$$[H(\omega)]^{-1} = \left\{ \begin{aligned} & -\omega^2 \begin{bmatrix} M_i + [\varphi_i(x_j)][m_{j-TMDI}][\varphi_i(x_j)]^T + [\varphi_i(x_j)(1 - \Delta\varphi_j)][b_j][\varphi_i(x_j)(1 - \Delta\varphi_j)]^T & [\varphi_i(x_j)][m_{j-TMDI}] + [\varphi_i(x_j)(1 - \Delta\varphi_j)][b_j] \\ [m_{j-TMDI}][\varphi_i(x_j)]^T + [b_j][\varphi_i(x_j)(1 - \Delta\varphi_j)]^T & [m_{j-TMDI}] + [b_j] \end{bmatrix} \\ & + j\omega \begin{bmatrix} 2M_i\omega_{ni}\xi_{ni} - 2M_i\omega_{ni}\xi_{ae} & \\ & 2m_{j-TMDI}\omega_{j-TMDI}\xi_{j-TMDI} \end{bmatrix} + \begin{bmatrix} M_i\omega_{ni}^2 & \\ & m_{j-TMDI}\omega_{j-TMDI}^2 \end{bmatrix} \end{aligned} \right\}^{-1} \quad (A8)$$

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